

COMP 3311: Database Management Systems

Lab 4 Exercise: Oracle Functions, SQL Aggregate Functions and Subqueries

WHAT TO DO

1. **Download** the zipped folder **Lab4ExerciseFiles.zip** from the *Lab 4* entry of the *Lab Assignments* section on Canvas to the desktop and unzip it. The folder contains two script files **lab4db.sql** and **lab4Queries.sql**. The **lab4db.sql** script file drops the **Student** and **AcademicUnit** tables previously created, if any, and creates four tables **AcademicUnit**, **Course**, **Student** and **Enrollsn**.
2. **Place** your **insertMyself.sql** script file inside the **Lab4ExerciseFiles** folder and **modify** it to also insert into the **Enrollsn** table five tuples with the following values. ← **NEW!**

studentId	courseId	grade
<your student id>	COMP3311	95.6
<your student id>	COMP4021	88.3
<your student id>	ELEC3100	93.1
<your student id>	HUMA1020	88.4
<your student id>	MATH2421	90.5

3. **Execute** the **lab4db.sql** script file in **SQL Developer**.
4. **Modify** the **lab4Queries.sql** script file by constructing SQL queries in the indicated locations in the script file.
5. **Test** your queries to ensure that they execute without errors.

EXPECTED QUERY RESULTS

MINIMUM	MAXIMUM	AVERAGE	MEDIAN	STUDENTS
38.2	99	75.85	82.4	21

Query 1 result.

UNITNAME	AVG CGA
Department of Mathematics	3.11
Department of Computer Science and Engineering	3.06
Department of Electronic and Computer Engineering	2.67
Department of Management	2.14

Query 2 result.

Note: The expected query result of the remaining queries will be presented in the next lab.

WHAT TO SUBMIT

Your modified **lab4Queries.sql** script file containing your SQL queries.

This is an individual exercise. Each student must submit their own SQL script file.

How To SUBMIT

By 6:00 p.m. tomorrow, upload your modified **lab4Queries.sql** script file to the *Lab 4 Assignment* on Canvas.

Note: You will only be able to submit a file with extension **sql**.

TEST YOUR SCRIPT FILE – USE ONLY SQL CONSTRUCTS DISCUSSED IN THE COURSE

The SQL queries in your submission will be tested by executing them against the **lab4db** database. If any query raises an SQL error or if you use SQL constructs not discussed so far in the course, then you will receive at most 0.5 marks for this exercise. Therefore, test your SQL queries against the **lab4db** database before you submit them and use only SQL constructs discussed so far in the course.